Economic Order Quantity Model and Extensions

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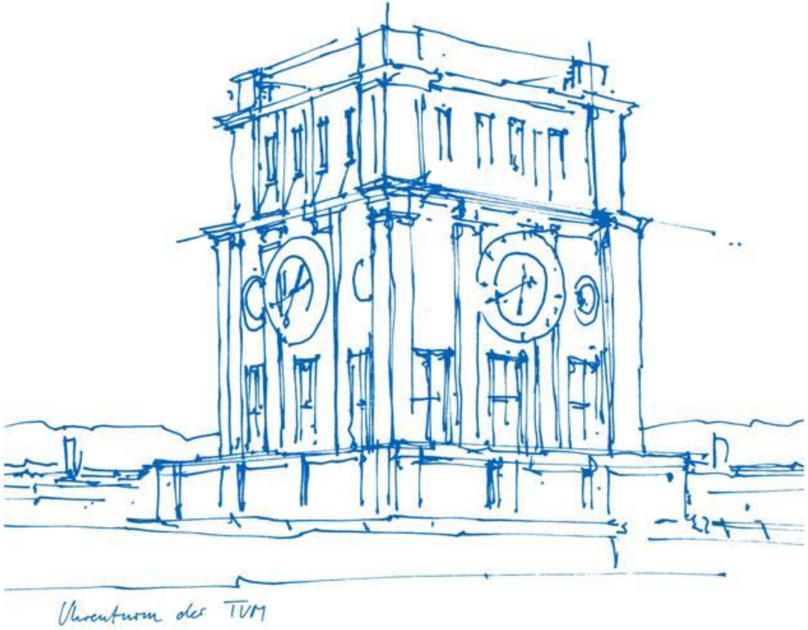
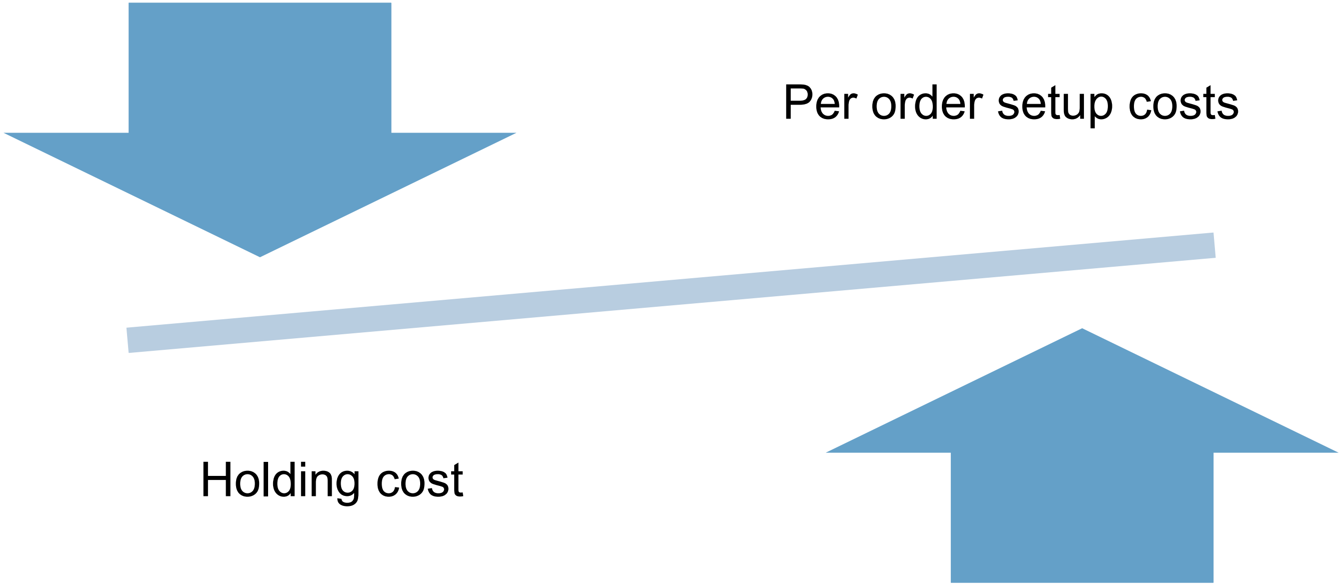


Basic Economic Order Quantity (EOQ) Model: Assumptions

Deterministic Inventory Management



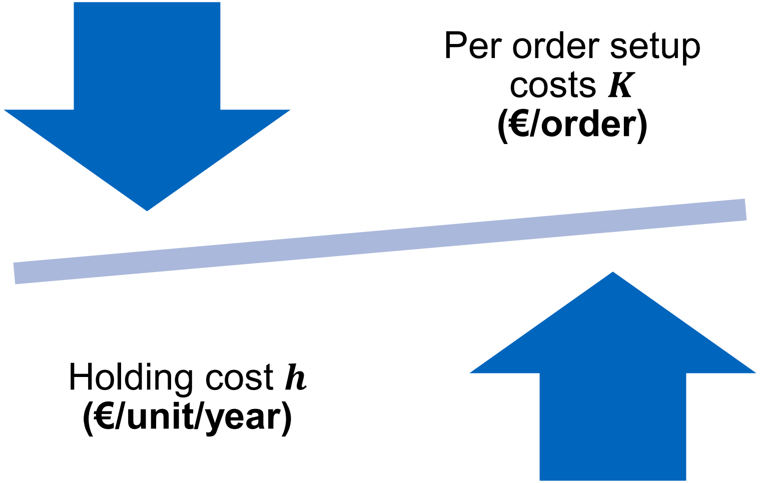
Deterministic Inventory Management

To find the optimal number of shipments (=orders) per year given deterministic and constant demand

- no initial inventory
- there is no order lead time
- infinite planning horizon
- only one product is involved
- the demand rate is constant and known
- shortages are not permitted
- costs include
  - setup cost
  - holding cost

Economic Order Quantity (EOQ)

Demand rate is constant and known: λ (units/year)



What is the optimal order quantity Q?

Basic Economic Order Quantity (EOQ) Model: Derivation

Inventory Levels for the EOQ Model

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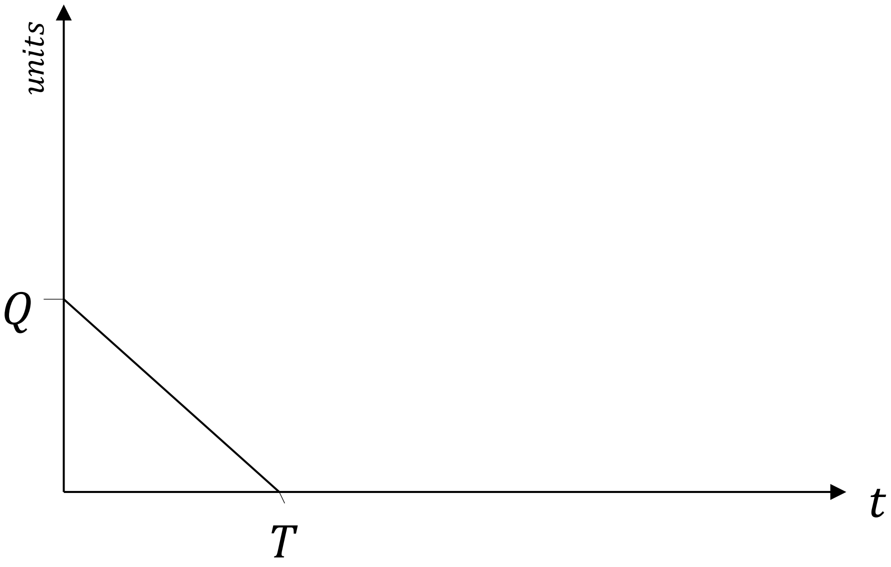
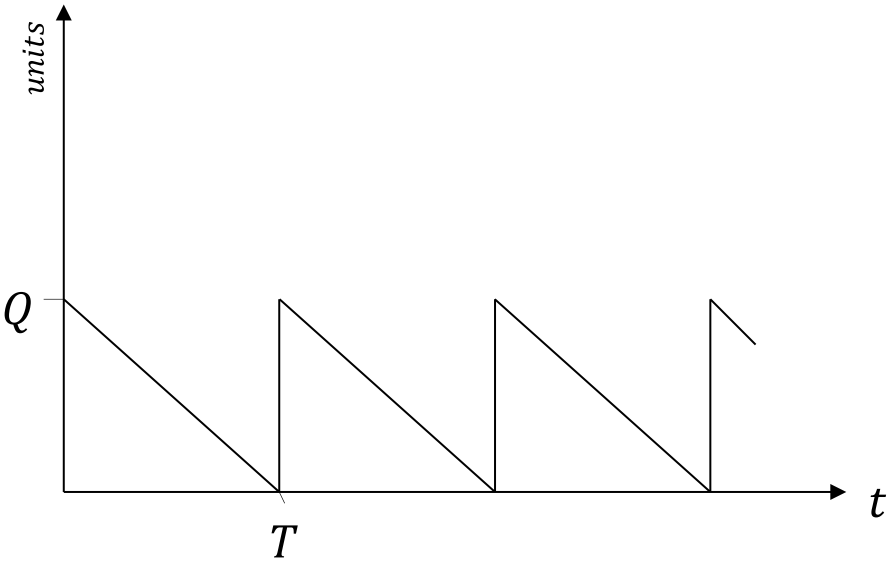
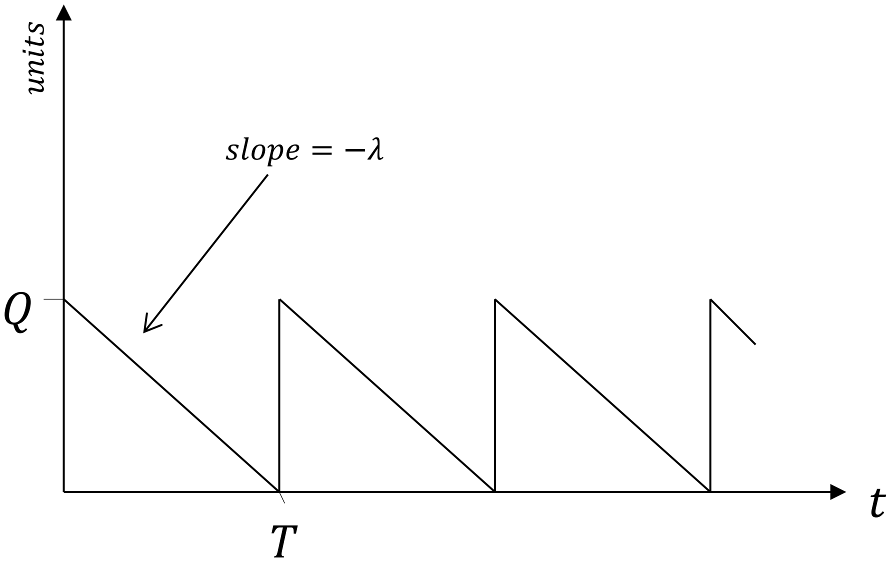


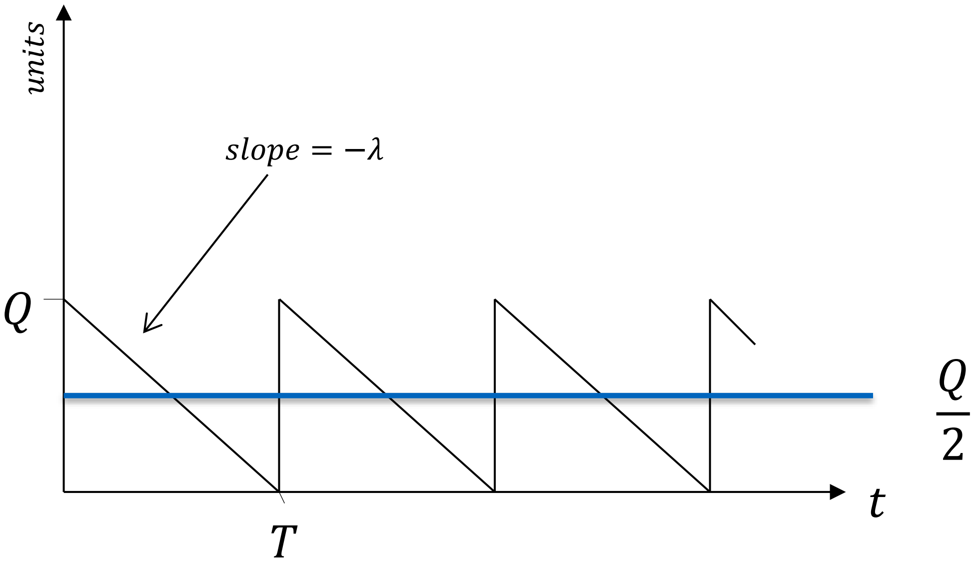
Inventory Levels for the EOQ Model



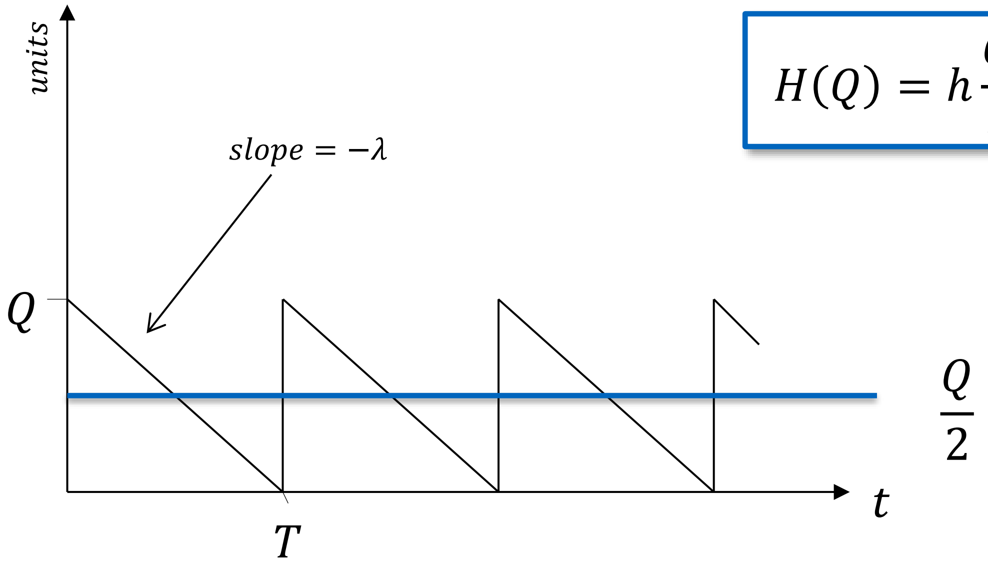
Inventory Levels for the EOQ Model



Inventory Levels for the EOQ Model



Inventory Levels for the EOQ Model



$$H(Q) = h\frac{Q}{2}$$

Average Number of Annual Orders

$$N = \frac{\lambda}{Q}$$

$$\boxed{k(Q) = K\frac{\lambda}{Q}}$$

Total Annual Costs

$$\boxed{TC(Q) = H(Q) + k(Q) = h\frac{Q}{2} + K\frac{\lambda}{Q}}$$

Optimal Order Quantity

$$TC'(Q) = \frac{h}{2} - K\frac{\lambda}{Q^2} := 0$$

$$\Rightarrow Q^* = \sqrt{\frac{2K\lambda}{h}}$$

$$TC''(Q) = 2K\frac{\lambda}{Q^3} > 0$$

$$\boxed{\mathrm{Q}^* =: EOQ = \sqrt{\frac{2K\lambda}{h}}}$$

Minimal Costs using the EOQ

$$TC(Q^*) = h\frac{Q^*}{2} + K\frac{\lambda}{Q^*} = h\frac{\sqrt{\frac{2K\lambda}{h}}}{2} + K\frac{\lambda}{\sqrt{\frac{2K\lambda}{h}}} = \sqrt{\frac{hK\lambda}{2}} + \sqrt{\frac{hK\lambda}{2}} = \sqrt{2hK\lambda}$$

$$\boxed{TC(Q^*) = \sqrt{2hK\lambda}}$$

Economic Order Quantity: Assumptions Revisited

Deterministic Inventory Management

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Positive Initial Inventory

Assume there is initial inventory, $I_0 > 0$.

<u>Implication for the EOQ model solution:</u> Wait I_0/λ and order amount Q^*. Then continue with the normal EOQ model.

Positive Order Lead Time

Assume it takes lead time l for an order to arrive (e.g., two weeks or one month).

Implication for the EOQ model solution: Each order cycle, order amount Q^* as before. But place the order when the inventory level is λl rather than 0. This way, the order arrives exactly when the inventory level is 0.

Economic Order Quantity: Finite-Planning Horizon

The Finite-Horizon Model: Assumptions

“infinite planning horizon” → What about seasonal products?

Assume finite horizon on interval $[0, t]$.

Assume a total of $m \geq 1$ orders is placed.

Assume no initial inventory and no left-over inventory at t.

The Finite-Horizon Model: Notation

For $1 \leq i \leq m-1$ define T_i as the time between the i^{th} and the $(i+1)^{th}$ order and define T_m as the time between the m^{th} order and t.

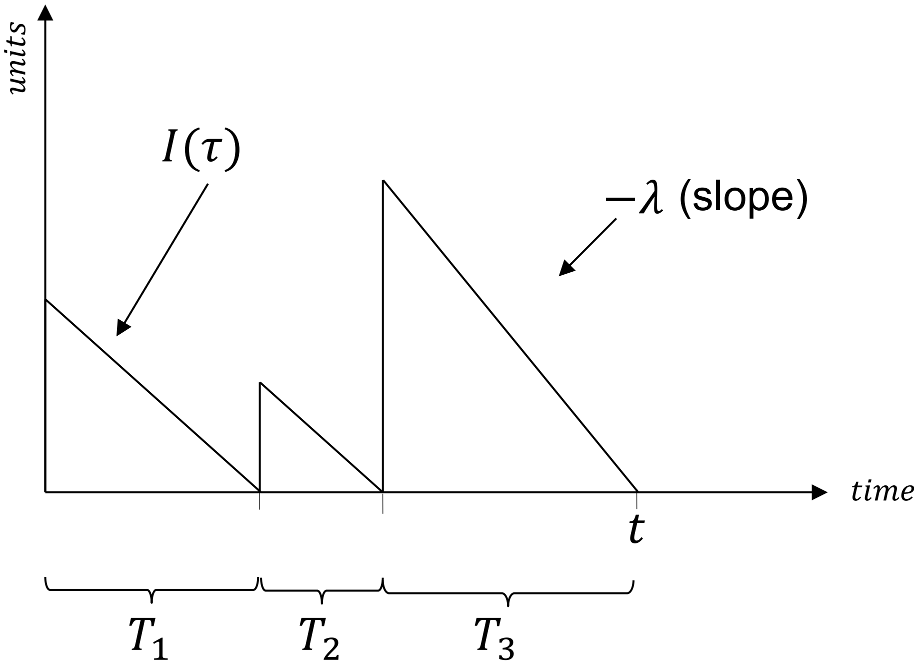
It follows:

$$\sum_{i=1}^{m} T_i = t$$

Let $\underline{t_i}$ denote the time at which order i arrives and $\overline{t_i}$ denote the time order order $i+1$ arrives if $i < m$ and the time t otherwise. That is, inventory related to order i is there only during the interval $\left[\underline{t_i}, \overline{t_i}\right]$.

Let $I(\tau)$ denote the inventory level at any time $\tau \in [0, t]$.

The Finite-Horizon Model: Inventory Levels Illustration



The Finite-Horizon Model: Optimization Problem

Normalize total cost to total costs per unit time,

$$
\begin{aligned}
TC &= \frac{1}{t}\left(Km + h\int_0^t I(\tau)\mathrm{d}\tau\right) \\
&= \frac{1}{t}\left(Km + h\sum_{i=1}^{m}\int_{\underline{t_i}}^{\overline{t_i}} I(\tau)\mathrm{d}\tau\right) \\
&= \frac{1}{t}\left(Km + h\sum_{i=1}^{m}\frac{T_i\,\lambda T_i}{2}\right) \\
&= \frac{1}{t}\left(Km + h\lambda\sum_{i=1}^{m}\frac{T_i^2}{2}\right)
\end{aligned}
$$

The Finite-Horizon Model: Minimization Logic

Problem:

$$min\left\{\sum_{i=1}^{m}\frac{T_i^2}{2}\,\middle|\,\sum_{i=1}^{m}T_i = t, T_i > 0\ \forall\ i = 1, \ldots m\right\}$$

Solution:

$$T_1 = T_2 = \cdots = T_m = \frac{t}{m}$$

The Finite-Horizon Model: Interim Solution

$$
\begin{aligned}
TC &= \frac{1}{t}\left(Km + h\lambda \sum_{i=1}^{m} \frac{\left(\frac{t}{m}\right)^2}{2}\right) \\
&= \frac{1}{t}\left(Km + \frac{h\lambda}{2m} t^2\right) \\
&= \frac{Km}{t} + \frac{h\lambda}{2m} t =: TC(m)
\end{aligned}
$$

This function is minimized with

$$
\widehat{m} = t\sqrt{\frac{h\lambda}{2K}}
$$

The Finite-Horizon Model: Integer Solution

$$TC(m) = \frac{Km}{t} + \frac{h\lambda}{2m} t$$

With

$$\hat{m} = t\sqrt{\frac{h\lambda}{2K}}$$

Total number of orders has the integer (!) and so the optimal solution m^* is

$$\lfloor \hat{m} \rfloor \text{ or } \lceil \hat{m} \rceil.$$

Check which one it is by plugging it in $TC(m)$. Order, each time, $Q^* = \frac{t\lambda}{m^*}$ units.

If $\widehat{m}$ is integer (and only in this case!)

$\widehat{m} = t\sqrt{\frac{h\lambda}{2K}}$ and $\mathrm{Q}^* = \frac{t\lambda}{m^*}$,

if $\widehat{m}$ = $\lfloor \widehat{m} \rfloor = \lceil \widehat{m} \rceil$ then $\widehat{m} = m^*$.

Therefore,

$$\mathrm{Q}^* = \frac{t\lambda}{t\sqrt{\frac{h\lambda}{2K}}} = \sqrt{\frac{2K\lambda}{h}}$$

…which is the standard EOQ solution!

The Finite-Horizon Model: Key Formulas

$$\widehat{m} = t\sqrt{\frac{h\lambda}{2K}}$$

$$m^* = \lfloor \widehat{m} \rfloor \text{ or } \lceil \widehat{m} \rceil$$

$$\mathrm{Q}^* = \frac{t\lambda}{m^*}$$

$$T_1 = T_2 = \cdots = T_m = \frac{t}{m}$$